

UNODC Laboratory and Scientific Service Portals

August 2025 - Emergence of potent kratom-related products containing 7-hydroxymitragynine and/or mitragynine pseudoindoxyl

VIENNA, Austria – 29 August 2025: Since 2024, novel [kratom](#)-related products have emerged on consumer markets, particularly in the United States. These products often contain high concentrations of [7-hydroxymitragynine](#) and/or [mitragynine pseudoindoxyl](#).^{1,2} Laboratory and surveillance methods need to adapt to monitor these more potent compounds.

The alkaloid [7-hydroxymitragynine](#) (7-OH) is a naturally occurring substance in the kratom plant (*Mitragyna speciosa*), but only as a minor constituent (formed during the drying process) compared with [mitragynine](#). [7-hydroxymitragynine](#) is also a metabolite of [mitragynine](#), and is further metabolized to [mitragynine pseudoindoxyl](#).²

Notably, [7-hydroxymitragynine](#) and [mitragynine pseudoindoxyl](#) demonstrate substantially greater mu-opioid receptor potency than [mitragynine](#). The risk of overdose is high, as [7-hydroxymitragynine](#) may be more potent than morphine at causing respiratory depression.^{1,3} Other health concerns include physical dependence and withdrawal symptoms similar to those caused by classical opioids.³

Recent testing of six novel kratom-related products marketed as “[7-hydroxymitragynine](#)” revealed that [7-hydroxymitragynine](#) was contained in all samples together with [mitragynine pseudoindoxyl](#), [mitragynine](#) and [paynantheine](#) in varying concentrations¹ (see photo 1). According to Hill and others, these novel “7-OH” products are likely synthesized in laboratories from purified [mitragynine](#).²

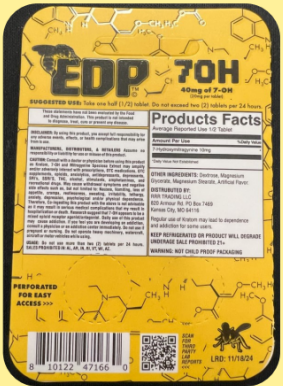
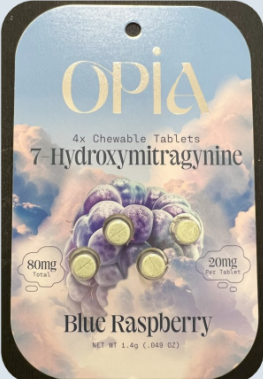
"7-HYDROXY MITRAGYNE" MARKETED PRODUCTS	LAB RESULTS
 	▶ 7-Hydroxy Mitragynine (1p) ▶ Mitragynine Pseudoindoxyl (0.3p) ▶ Mitragynine (0.2p) ▶ Paynantheine (0.1p) 
 	▶ 7-Hydroxy Mitragynine (1p) ▶ Mitragynine Pseudoindoxyl (0.03p) ▶ Mitragynine (0.03p) ▶ Speciogynine, Speciociliatine, & Mitraciliatine (all >0.01p) ▶ Paynantheine (0.1p) 

Photo 1: Examples of “7-hydroxymitragynine” marketed products and their lab results

Note: Parts designation is described by Krotulski and others as follows: “p” = parts. 0.5p = half as abundant as 1p. 2p = 2x more abundant than 1p.¹

Source: Krotulski and others.¹

The growing number of semi-synthetic kratom-related products are found in diverse forms including chewable/sublingual tablets or strips, gummies, drink mixes, shots, syrups, vape pens, capsules, food products or powders.² The marketing strategies and easy accessibility of these products raise significant health concerns, as these products may not be clearly or accurately labelled², might be designed to appeal to young people (see photo 2) and are available in corner stores, gas stations and vape shops in the United States.^{4,5} Moreover, “7-hydroxymitragynine” marketed products are also available on websites outside of the United States.*



Photo 2: Forms and design of new kratom-related (7-hydroxymitragynine) products

Source: United States, Food and Drug Administration.⁶

Hill and others compiled a database with more than 304 [7-hydroxymitragynine](#) and [mitragynine pseudoindoxyl](#) marketed products (not chemically verified). Most products claimed to be “kratom-based” and suggested effects such as “*general wellbeing*”, “*pain relief*” or “*stress/anxiety reduction*”, misleading consumers who intend to use a “low-potency botanical supplement” but instead end up with an opioid-like NPS.²

So far, since 2024, the United States and Australia have reported toxicology cases with high concentration [7-hydroxymitragynine](#) products being involved to the [UNODC Early Warning Advisory \(EWA\) on New Psychoactive Substances Tox-Portal](#) (data collection is ongoing).⁷

[Kratom](#), [mitragynine](#), [7-hydroxymitragynine](#) and [mitragynine pseudoindoxyl](#) are currently not controlled under the UN conventions⁸. In the United States, [kratom](#) and its constituents are not nationally controlled.³ The United States Food and Drug Administration (FDA) state in their 7-OH assessment that “none of these substances are lawful when added to conventional foods, as dietary supplements, or as ingredients in any FDA-approved drug”.³ In July 2025, FDA recommended a scheduling action to control certain [7-hydroxymitragynine](#) products under the Controlled Substances Act.⁴ Some countries control [kratom](#) and/or [mitragynine](#) (e.g. [Australia](#), [Slovakia](#)), whilst others have specifically additionally controlled [7-hydroxymitragynine](#) and/or other kratom alkaloids (e.g. [France](#), [Luxembourg](#), [Singapore](#)).

Response options

- **Raise awareness among:**
 - + **health care providers** (e.g. see [FDA letter for health care professionals](#); [Pennsylvania Department of Health alert](#)),
 - + **law enforcement**,
 - + **regulators** (consider potential gaps in existing legislation),
 - + **people who use drugs and/or the public**, if locally relevant (e.g. see [FDA educational material](#)⁶)
- **Ensure access to lifesaving treatments such as naloxone**³
- **Provide comprehensive drug dependence treatment and care programmes** as important components of **overdose prevention**
- **Enhance capabilities of forensic drug testing and toxicology laboratories** - reference standards and appropriate analytical methods are needed to effectively identify and differentiate [mitragynine](#)-related compounds. Complexities such as metabolism³, poor stability especially in biological samples^{1,9} and limitations of some standard methods such as GC-MS need to be considered.¹
- **Include [7-hydroxymitragynine](#) and [mitragynine pseudoindoxyl](#) in data collections** for early warning systems, drug observatories and (self-reported) drug use surveys.³ [UNODC EWA](#) has responded already by creating NPS entries and analytical findings are encouraged to be shared.

 * Results of a Google search using “7-OH”, “buy”, “comprar”, “kaufen” and/or the respective country name.

****We would like to acknowledge the contributions from the United States Drug Enforcement Administration and the Emerging Drugs Network of Australia to the [UNODC EWA Tox-Portal](#).*

References:

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